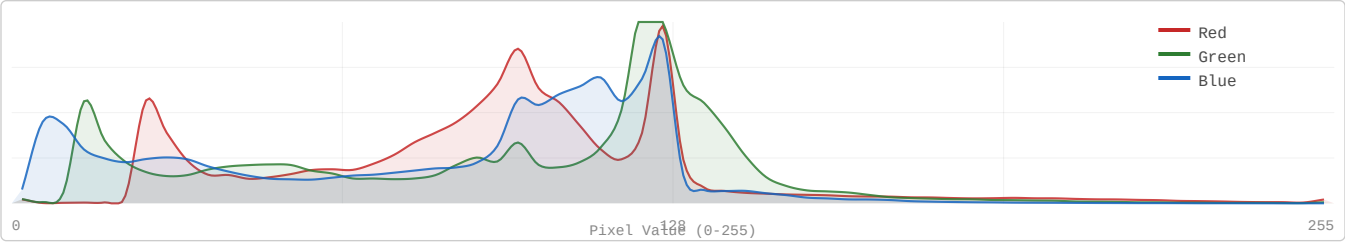
Decorrelation Gap: KODIM11

768x512 | 24-bit RGB | PNG (lossless)

Lakeside Cabin in Mountains | Weakly one-dimensional (PC1 = 91.56%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.8225	0.7906
G	0.8225	1.0000	0.9744
B	0.7906	0.9744	1.0000

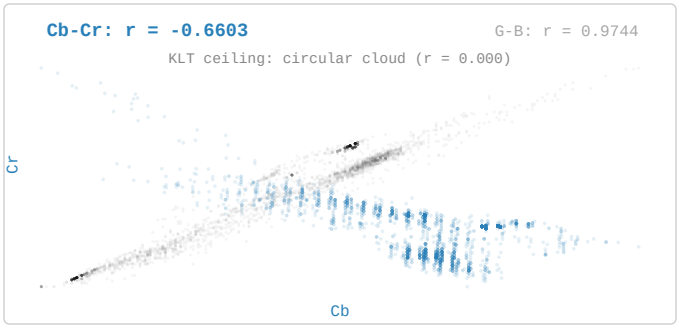
Cb-Cr: -0.660

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.0166	-0.3167
Cb	-0.0166	1.0000	-0.6603
Cr	-0.3167	-0.6603	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.822536	-0.016647	0.0166
R-B → Y-Cr	0.790610	-0.316690	0.3167
G-B → Cb-Cr	0.974415	-0.660333	0.6603
Avg |r|	0.862520	0.331223	0.3312

Decorrelation Gap: KODIM11

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Lakeside Cabin in Mountains | Weakly one-dimensional (PC1 = 91.56%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

61.6% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

5.0%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	43.46 dB
G	48.82 dB
B	44.43 dB
Overall	45.03 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

PC1	91.56%
PC2	7.59%
PC3	0.85%

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5047	-0.6286	-0.5918	5021.44
PC2	-0.8596	0.3027	0.4116	416.03
PC3	0.0796	-0.7164	0.6931	46.67

8. PCA Derived Metrics

Condition Number

107.60

PC1/PC2: 12.07 | PC2/PC3: 8.91

Blue Channel Independence

5.0%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.63)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 107.60 | PC1: 91.6%

RGB Correlation: Avg |r| = 0.862520 | R-G: 0.8225 | R-B: 0.7906 | G-B: 0.9744

YCbCr Residual: Avg |r| = 0.331223 | Y-Cb: -0.0166 | Y-Cr: -0.3167 | Cb-Cr: -0.6603

Efficiency: 61.6% of KLT ceiling | Gap: 0.3312 avg |r| remaining

Blue Channel: 5.0% independent | 4:2:0 PSNR: 45.03 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.